

PYTHON PROJECT

HOUSE PRICE PREDICTION

Introduction:

House price prediction is an important application of data analysis and machine learning in the real estate domain. It involves estimating the selling price of a property based on various factors such as locations, physical characteristics, condition of the house and other relevant features. Accurate prediction of the house prices helps buyers, sellers and real estate agents make informed decisions.

In this project we analyse a real estate dataset containing both numerical and categorical features related to residential properties. The primary objective is to understand how different factors influence house prices and to build a predictive model that can estimate the sale price of a house.

The dataset includes features such as lot area, zoning classification, building type, overall condition, year of construction, year of remodelling, basement size, and exterior materials. These features are analysed to identify their relationship with target variable, which is the SalePrice.

The project follows a structured approach that includes data exploration, data cleaning, handling missing values, feature engineering, and visualisation. Various statistical and graphical techniques are used to uncover patterns and the insights from the data.

Finally, a machine learning model is developed to predict house prices based on the processed features. The performance of the model is evaluated using appropriate

metrics, ensuring that the predictions are reliable and meaningful.

Methodology:

1. Data Collection and Loading:

The dataset was provided as part of the project and loaded using Python libraries such as Pandas. It contains various features related to house properties along with the target variable, SalePrice.

2.Data Understanding and Exploration

Initial exploration performed to understand the structure of the dataset.

This included:

- Viewing column names and data types
- Displaying sample records
- Understanding the meaning of each feature

This step helped in identifying numerical, categorical and temporal variables.

3.Data Preprocessing:

Handling missing values: Missing values in categorical features were filled using mode

Dataset Separation: The dataset contained both training and testing data. Rows with missing SalePrice values were separated as test data, and only in training data was used for model building.

4.Exploratory Data Analysis:

- Scatter plots were used for continuous variables
- Box plots were used for discrete variables
- Bar charts were used for categorical variables
- Histogram were used to study data distribution

5.New Features:

- House Age: Derived from YearBuilt
- Remodel Age: Derived from YearRemodAdd
- Categorical variables were converted into numerical form using One-Hot Encoding.

6.Data Transformation:

Logarithmic transformation was applied to the SalePrice variable to reduce skewness and improve the models performance

7.Feature Selection:

Correlation analysis was performed to identify features that have a strong relationship with SalePrice. Relevant features were selected for model training.

Requirement Analysis:

1.Software Requirements:

Programming Language used is Python

The development environment is Google Colab

The libraries used are Pandas(Data Manipulation), NumPy(for numerical computations), Matplotlib & Seaborn(for data visualisation)

2.Hardware Requirements:

A computer system with a minimum of 4GB RAM

Processor: Intel i3 or above

3.Dataset Requirements:

The dataset contains both training and testing data related to house properties

It includes numerical, categorical and temporal features.

The target variable is SalePrice, which is used for prediction.

4.Functional Requirements:

Load and preprocess the dataset

Handle missing values appropriately

Perform EDA

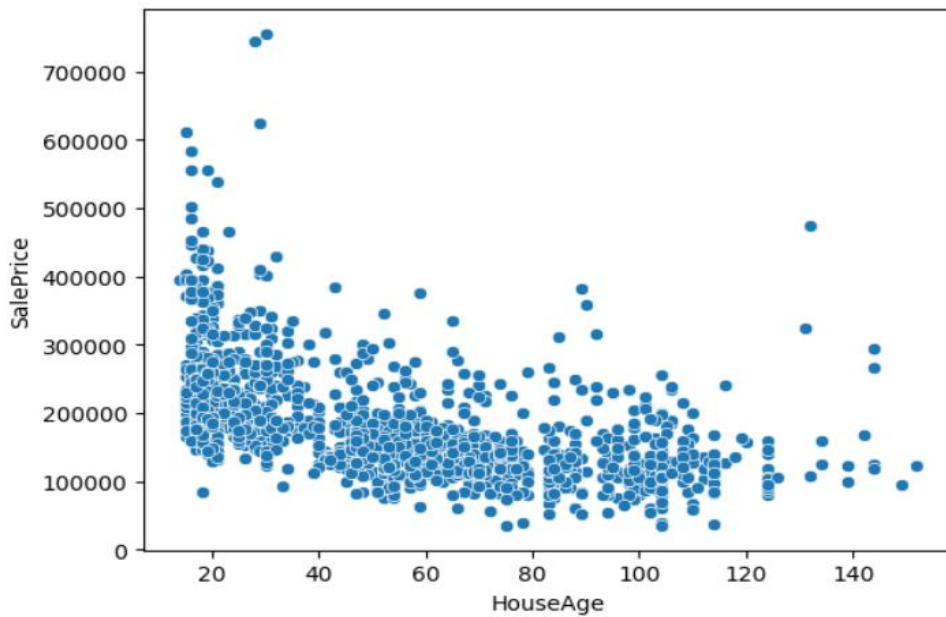
Apply feature engineering techniques

Evaluate model performance using appropriate metrics

Predict house prices for unseen data.

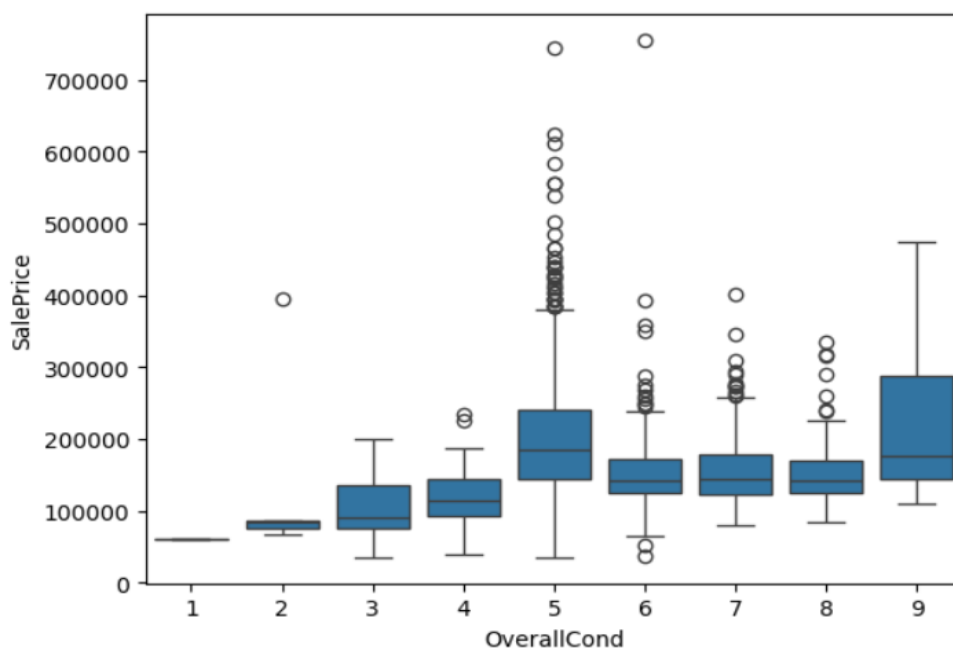
Visualisation and Insights:

1.House Age vs SalePrice



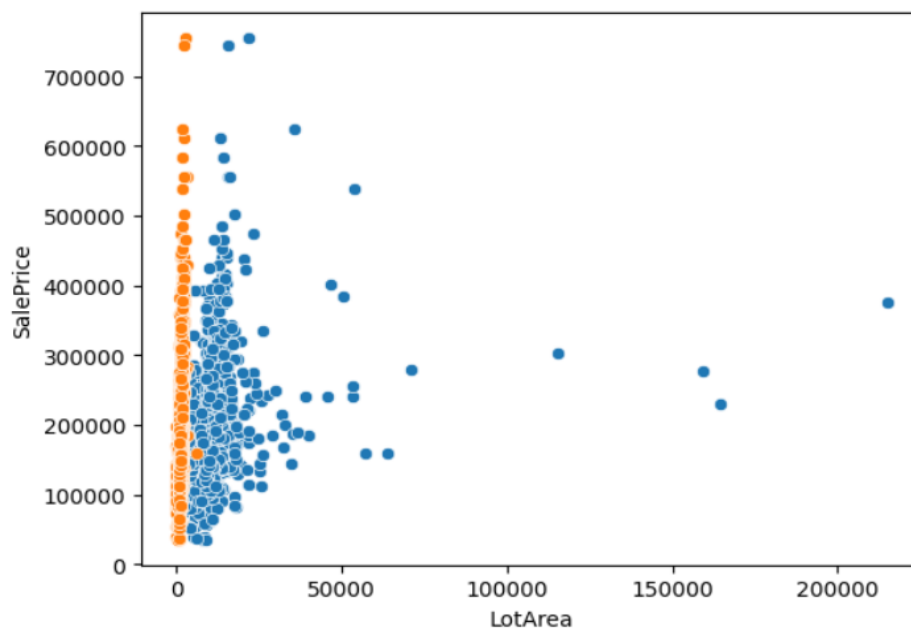
Insight: The scatter plot shows a clear negative relationship between house age and sale price. As the age of the house increase, the sale price tends to decrease. Newer houses are generally priced higher. However, some older houses still show higher prices.

2. Overall condition vs SalePrice(Box Plot)



Insight: The box plot indicates that houses with higher overall condition ratings tend to have higher sale prices. Properties with condition values around 5-7 show a significant increase in price compared to lower-rated house. This demonstrates that the physical condition of a house plays a crucial role in determining its market value.

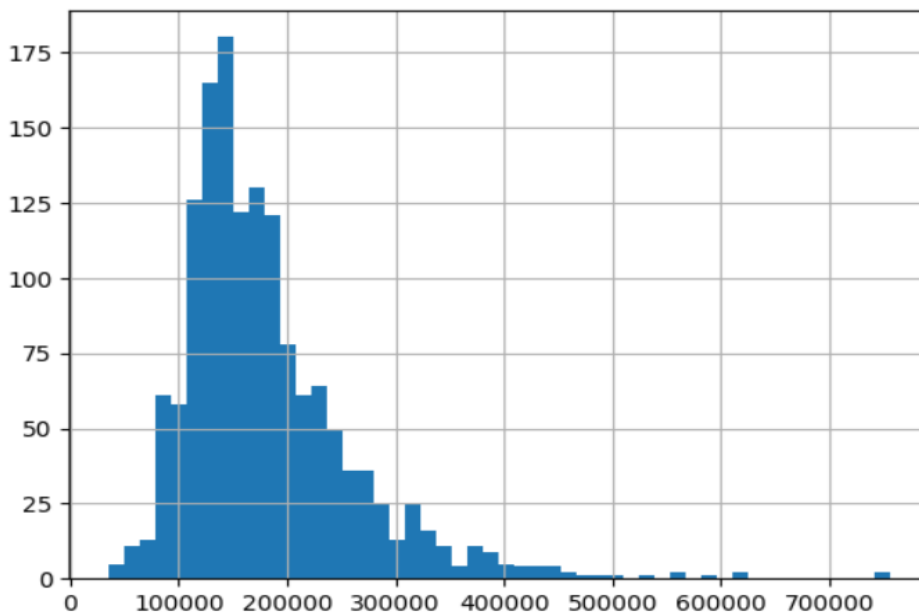
3. Lot Area vs SalePrice (Scatter Plot)



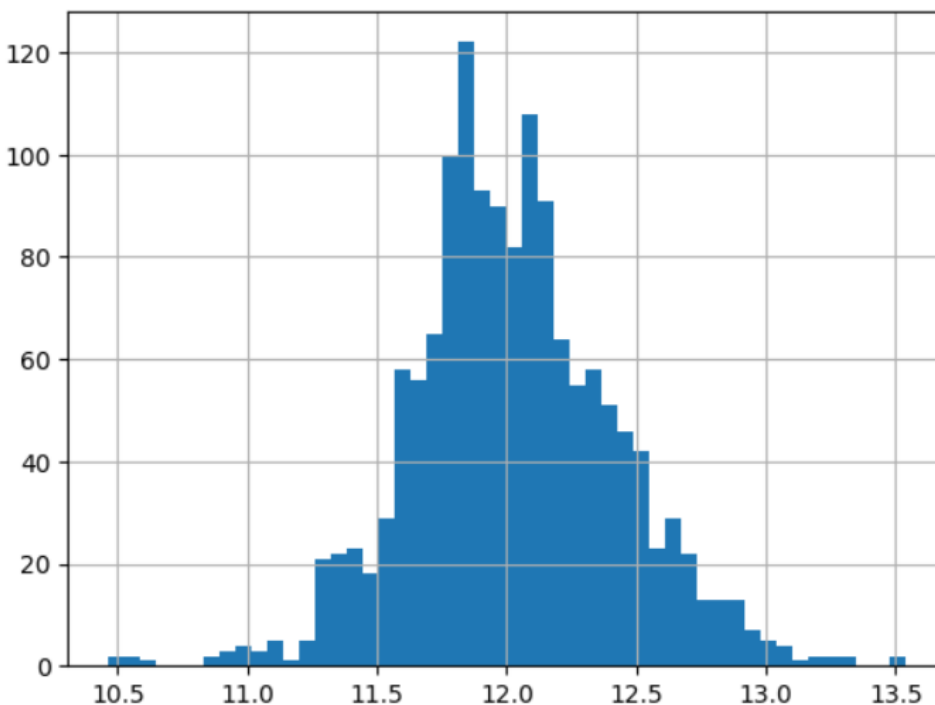
Insight: The relationship between lot area and sale price is moderately positive, Houses with larger lot areas generally have higher prices, but the relationship is not very strong due to the presence of outliers. Many houses with smaller lot

areas still have high prices, suggesting that location and other features also influence pricing.

4. SalePrice Distribution (Histogram)



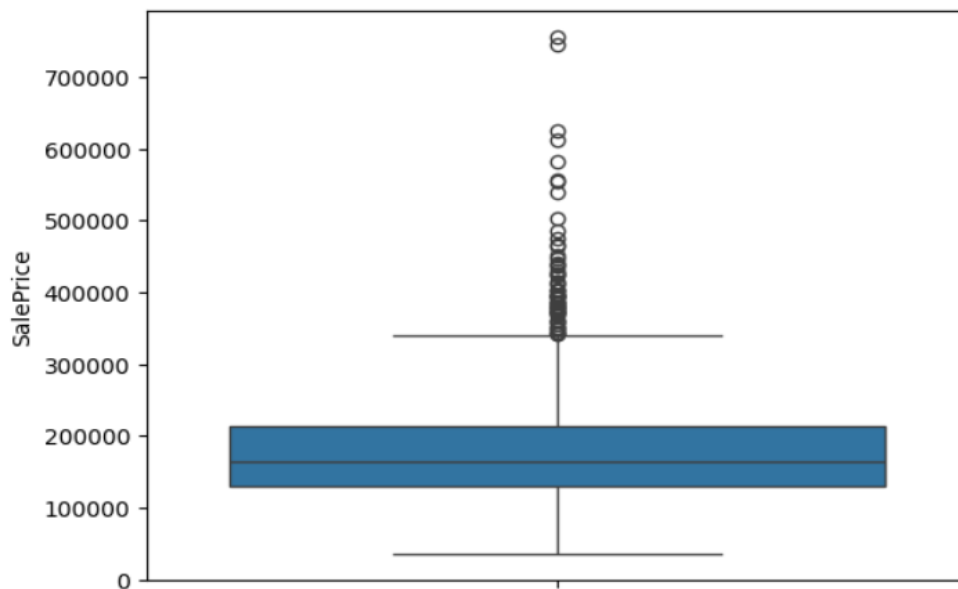
5. Log-Transformed SalePrice Distribution (Histogram)



Insight: After applying log transformation, the distribution becomes approximately normal. This improves the

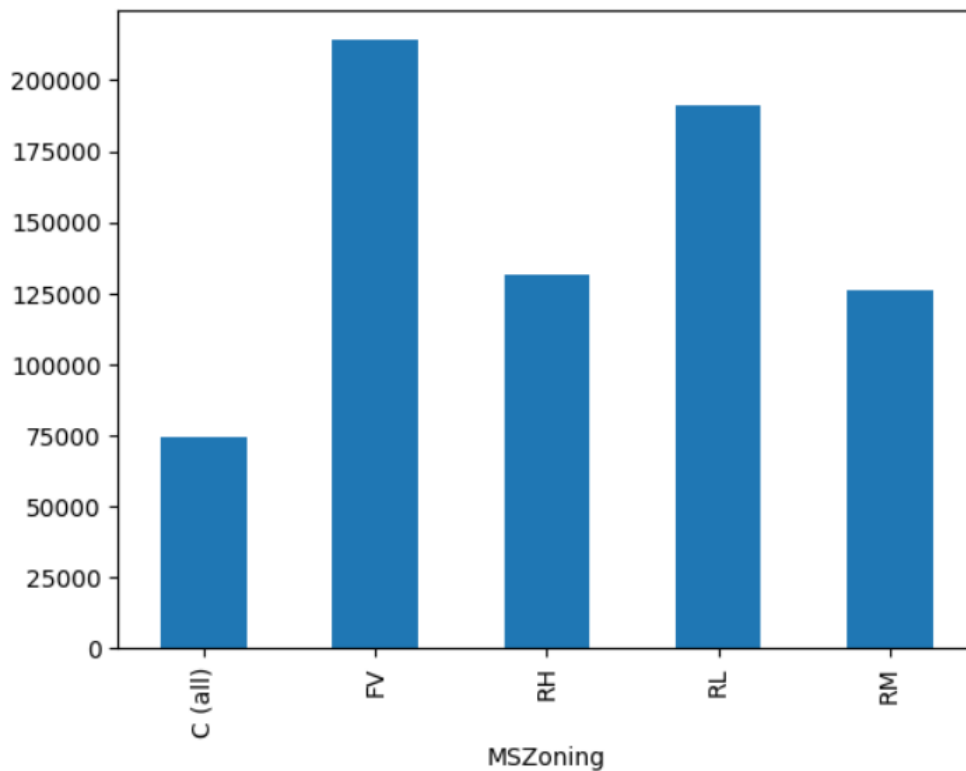
performance of machine learning models by stabilizing variance and reducing the impact of extreme values.

6. Outlier Detection (Box Plot of SalePrice)



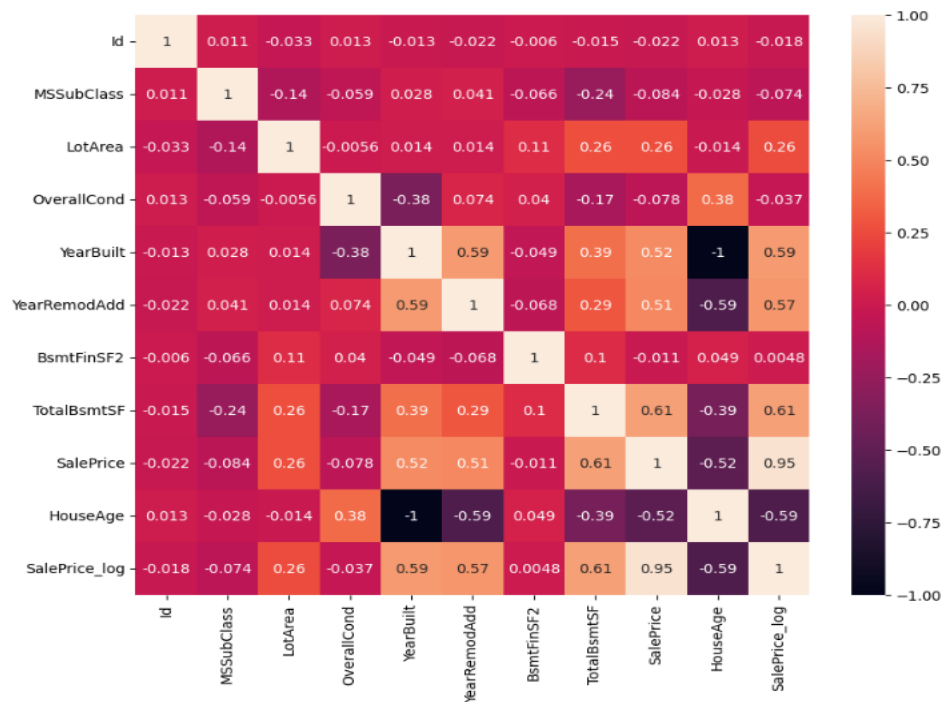
Insight: Zoning classification has a strong impact on house prices. Houses in premium residential zones such as a FV and RL have higher average prices, while properties in commercial zones have the lowest prices.

7. MSZoning vs SalePrice (Bar Chart):



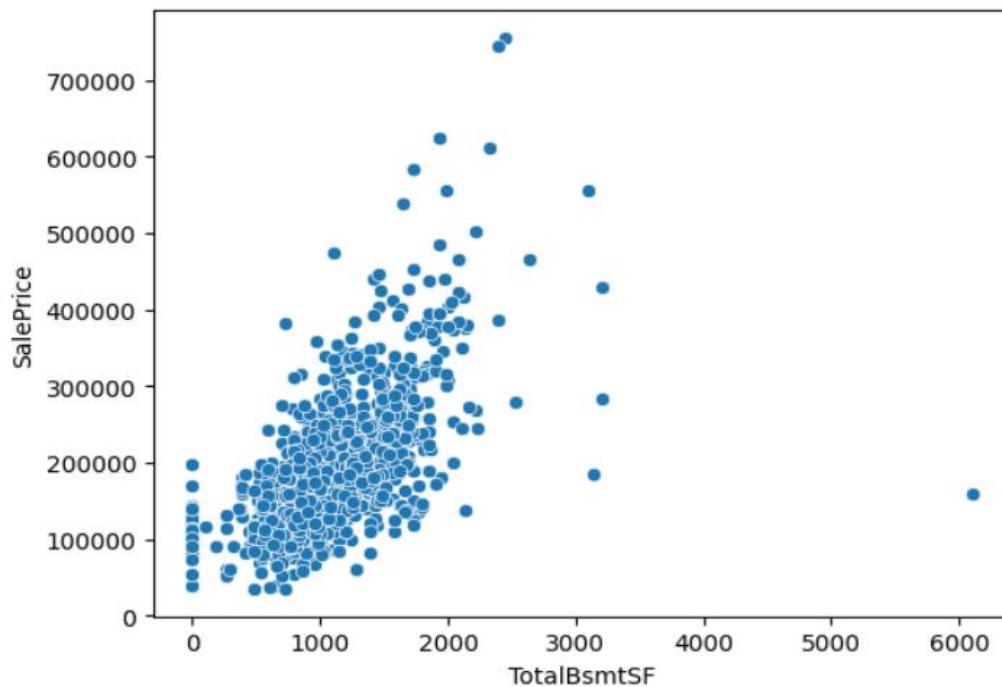
Insights: Zoning classification has a strong impact on house prices. Houses in premium residential zones such as FV and RL have higher average prices, while properties in commercial zones have the lowest prices. This highlights the importance of location in real estate valuation.

8. Correlation Heatmap:



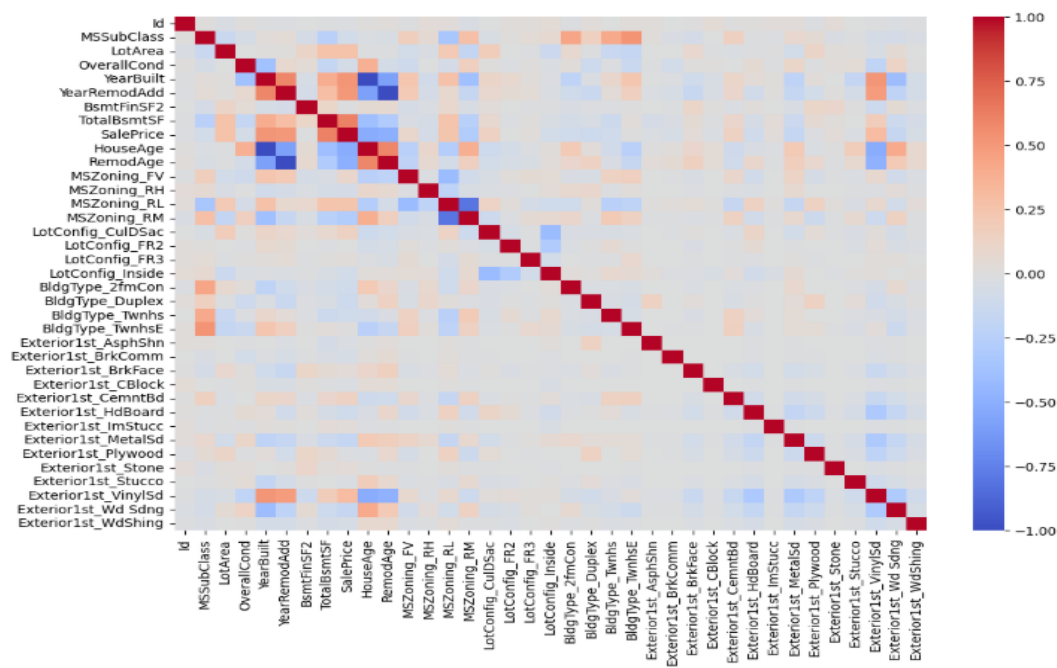
Insight: : The heatmap shows that features such as TotalBsmtSF, YearBuilt and YearRemodAdd have strong positive correlation with sale price. HouseAge shows a negative correlation, indicating that older houses tend to have lower prices. These features are important predictors for the model.

9. YearRemodAdd vs SalePrice (Scatter Plot)



Insight: The plot shows that recently remodeled houses tend to have higher sale prices. Renovation increases property value by improving aesthetics, functionality and overall appeal.

10. Full Feature Correlation Matrix



Insight: The correlation matrix after encoding shows relationships between multiple features. It confirms that both numerical and categorical variables contribute to price prediction. Some categorical variables also show meaningful relationships after encoding.

Conclusion:

In this project, we analysed different factors that influence house prices and from the analysis it was clear that features like location, house condition, basement size and age have a strong impact on pricing.

The data was cleaned and missing values were handles and in addition to that useful features like age was create to improve the results. The visualisations helped in understanding important patterns, such as newer and well- maintained houses having higher prices.

Overall, this project shows how data analysis can be applied to solve real world problems like this.